Reducing Employee Attrition: Findings and Recommendations

Oct 24, 2026 · @marienel

Problem

237 of 1,470 employees left, an attrition rate of 16.1%, which cost an estimated $6.8M in replacement costs. This memo shows who leaves, why, and who is likely to leave next, so HR can act before the resignations arrive.

Replacement cost is assumed at 50% of annual salary per leaver.

Key findings

Five groups leave at 1.8 to 2.5 times the company average, and overtime is the clearest single signal: 30.5% attrition with overtime against 10.4% without, 2.9 times higher.

Group	Attrition rate	Leavers
Company average	16.1%	237 of 1,470
Sales Representatives	39.8%	33 of 83
Age 18 to 25	35.8%	44 of 123
0 to 1 years at the company	34.9%	75 of 215
Works overtime	30.5%	127 of 416
Lowest pay quartile	29.3%	108 of 369

These groups overlap: younger, newer and lower-paid employees are often the same people.

Prediction

A logistic regression model identifies likely leavers with a ROC-AUC of 0.80 on employees it had not seen, and 97 current employees fall in its highest-risk band.

The model's three strongest drivers are overtime, years at the company, and frequent business travel. A random forest scored 0.79, so the simpler and more explainable model was kept. The 97 employees are listed with their risk scores on the dashboard's Flight Risk page.

Recommendations

Five actions, each aimed at a group that leaves well above the 16.1% average.

1. **Audit and cap overtime.** Review overtime hours in the roles with the highest attrition and redistribute work or add headcount. Owner: operations leads with HR business partners. Measure: attrition among employees who work overtime, currently 30.5%.
2. **Fix the first year.** Add check-ins at 30, 60 and 90 days and assign each new hire a buddy. Owner: HR and line managers. Measure: attrition at 0 to 1 years, currently 34.9%.
3. **Review the Sales Representative role.** Look at pay, targets and career path, and run stay interviews. Owner: sales director with compensation. Measure: role attrition, currently 39.8%.
4. **Benchmark pay in the lowest quartile.** Compare against market rates and correct gaps. Owner: compensation team. Measure: attrition in the lowest pay quartile, currently 29.3%.
5. **Hold stay conversations with the 97 high-risk employees within 30 days.** Owner: managers with HR business partners. Measure: how many of the 97 are still employed after six months.

Expected impact

Keeping 20% of the 97 high-risk employees would save about $613,664 in replacement costs.

If all 97 left, replacement would cost about $3.07M. Every extra 10 percentage points of retention adds roughly $307,000 in savings. The 20% retention rate is an assumption to test in a pilot, not a forecast.

Limitations and assumptions

These results show patterns in one public sample dataset, so treat them as a method demonstration rather than proof for any real company.

- The data is IBM's public sample of 1,470 employees, not a live company's records.
- The patterns are correlations. They show who leaves, not why.
- Groups overlap, so the effects of age, tenure and pay should not be added together.
- Some groups are small: only 83 employees are Sales Representatives.
- The 50% replacement cost and 20% retention rate are assumptions.
- The model used age, gender and marital status. Before real use, remove them and test the model for bias, so no decision about a person rests on a protected characteristic.